

02_null_experiment

February 16, 2026

1 02 — Null experiment

Purpose: For each dataset and seed, compare observed Moran's I to the null distribution (permuted predictions), plot null histograms with observed line, report empirical p-values, and summarize fraction of significant runs.

1.1 Imports and path setup

```
[5]: import sys
from pathlib import Path

ROOT = Path.cwd().parent if Path.cwd().name == "notebooks" else Path.cwd()
if str(ROOT) not in sys.path:
    sys.path.insert(0, str(ROOT))
if str(ROOT / "src") not in sys.path:
    sys.path.insert(0, str(ROOT / "src"))

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from analysis.experiment_runner import _get_run_dirs
from analysis.preprocessing import get_transformed_test_features
from analysis.run_analysis import load_config, load_meta, run_spatial, run_null

RESULTS_DIR = ROOT / "results"
```

1.2 Load run artifacts and run null experiment

For each run we get preprocessed X_{test} via `analysis.preprocessing.get_transformed_test_features(run_dir, dataset_name)` so preprocessing matches training exactly. Then call `run_spatial` and `run_null` to get observed Moran's I and null distribution.

```
[6]: DATASET = "compas"
K = 25
R_null = 100
k_nn = 30
```

```

dataset_dir = RESULTS_DIR / DATASET
run_dirs = _get_run_dirs(dataset_dir)
if not run_dirs:
    raise SystemExit(f"No run dirs in {dataset_dir}")

run_results = []

for run_dir in run_dirs:
    config = load_config(run_dir)
    outer_seed = config.get("outer_seed", int(run_dir.name.split("=")[1]) if
↪ "=" in run_dir.name else 0)
    n_cand = len(load_meta(run_dir))
    K_actual = min(K, n_cand)
    X_test = get_transformed_test_features(run_dir, DATASET)
    spatial = run_spatial(run_dir, X_test, K=K_actual, k=k_nn)
    null = run_null(run_dir, X_test, K=K_actual, R=R_null, k=k_nn)
    run_results.append({
        "outer_seed": outer_seed,
        "observed_moran": spatial["moran_i"],
        "null_moran_i": null["null_moran_i"],
        "p_empirical": (1 + np.sum(null["null_moran_i"] >= spatial["moran_i"]))
↪ / (R_null + 1),
    })

run_results = pd.DataFrame(run_results)
run_results

```

```

[6]:      outer_seed  observed_moran  \
0           0         0.140503
1           1         0.211425
2           2         0.205800
3           3         0.360500
4           4         0.110097
5           5         0.140545
6           6         0.175005
7           7         0.215078
8           8         0.291669
9           9         0.142169

```

```

                                null_moran_i  p_empirical
0  [-0.007014052948101744, -0.0002955497401562206...  0.009901
1  [-0.0030747869382435145, 0.0019983039357006366...  0.009901
2  [-0.0030272200746984195, 0.0031243212354898314...  0.009901
3  [0.00028620482932572173, -0.004133374222726578...  0.009901
4  [-0.004468562807067691, 0.004327043460574827, ...  0.009901
5  [-0.003855403535809964, -0.011456184226235025,...  0.009901
6  [0.007003053729409927, -0.0024286318378795332,...  0.009901

```

```

7 [0.003032179243585836, 0.0024188572398711203, ... 0.009901
8 [-0.006001903642729862, -0.007010142039237414,... 0.009901
9 [0.004319151346286712, -0.007969756215970469, ... 0.009901

```

1.3 Visualization: null distribution histograms with observed line

For each run (or a subset), plot histogram of null Moran's I and a vertical line at observed Moran's I.

```

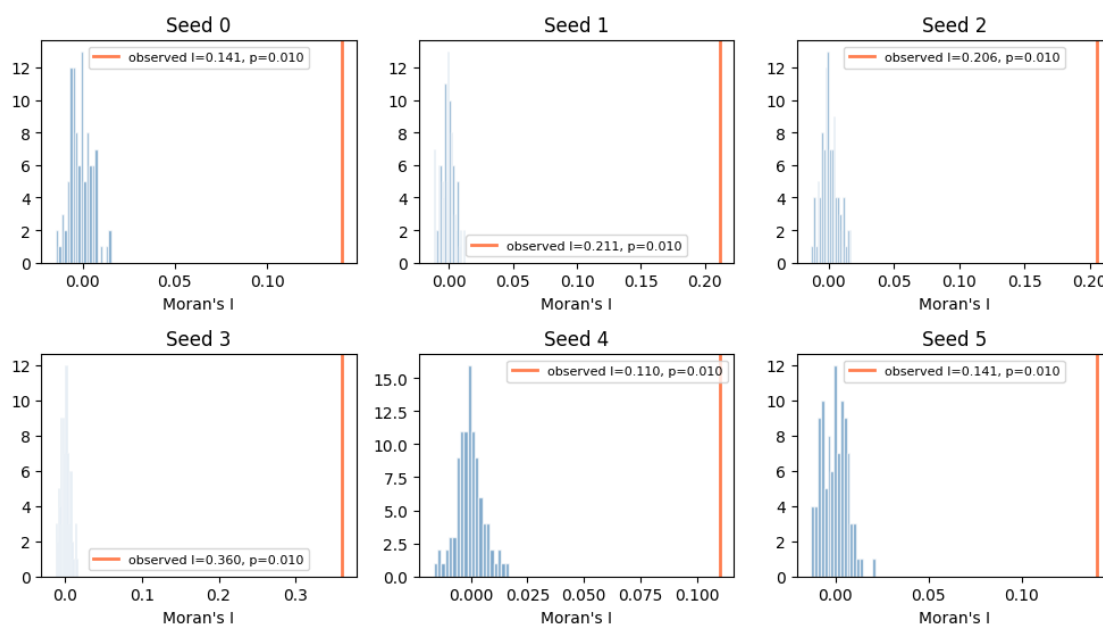
[7]: n_show = min(6, len(run_results)) # show first 6 runs
fig, axes = plt.subplots(2, 3, figsize=(10, 6))
axes = axes.flatten()

for i, (_, row) in enumerate(run_results.head(n_show).iterrows()):
    ax = axes[i]
    null_i = row["null_moran_i"]
    obs = row["observed_moran"]
    p = row["p_empirical"]
    ax.hist(null_i, bins=20, alpha=0.7, color="steelblue", edgecolor="white")
    ax.axvline(obs, color="coral", lw=2, label=f"observed I={obs:.3f}, p={p:.3f}")
    ax.set_title(f"Seed {row['outer_seed']}")
    ax.legend(fontsize=8)
    ax.set_xlabel("Moran's I")

for j in range(i + 1, len(axes)):
    axes[j].set_visible(False)
plt.suptitle(f"Null distribution of Moran's I ({DATASET}, R={R_null})")
plt.tight_layout()
plt.show()

```

Null distribution of Moran's I (compas, R=100)



1.4 Aggregated significance table

Fraction of runs with significant Moran's I ($p_{\text{empirical}} < 0.05$).

```
[8]: run_results["significant"] = run_results["p_empirical"] < 0.05
frac_sig = run_results["significant"].mean()
print(f"Dataset: {DATASET}")
print(f"Runs: {len(run_results)}")
print(f"Fraction significant ( $p_{\text{empirical}} < 0.05$ ): {frac_sig:.2%}")
display(run_results[["outer_seed", "observed_moran", "p_empirical",
    ↪ "significant"]])
```

Dataset: compas

Runs: 10

Fraction significant ($p_{\text{empirical}} < 0.05$): 100.00%

	outer_seed	observed_moran	p_empirical	significant
0	0	0.140503	0.009901	True
1	1	0.211425	0.009901	True
2	2	0.205800	0.009901	True
3	3	0.360500	0.009901	True
4	4	0.110097	0.009901	True
5	5	0.140545	0.009901	True
6	6	0.175005	0.009901	True
7	7	0.215078	0.009901	True
8	8	0.291669	0.009901	True
9	9	0.142169	0.009901	True